

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

US-based (Kent, OH), F-1 OPT, fully remote-ready

PROFESSIONAL SUMMARY

Backend / AI Platform engineer (2.5+ years) who ships production Python services for LLM and ML systems. Built FastAPI/Flask RESTful inference services, containerized with Docker, with structured logging and load simulation, cutting post-deployment defects ~30%. Strong with PostgreSQL and relational data modeling, RAG and LLM application development, and cloud-ready deployment. Comfortable owning backend services and APIs end to end and growing into conversational-AI platform work at scale.

CORE COMPETENCIES

Python • FastAPI / Flask • REST APIs • PostgreSQL / SQL • Docker • Cloud-ready deployment • LLM application development • RAG / embeddings • Observability / logging • CI/CD

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS

- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Built **Retrieval-Augmented Generation (RAG)** systems with recursive semantic chunking and transformer embeddings; improved retrieval precision ~35%, grounding alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved response stability ~25%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts and support logs; raised dataset reliability above **98%** and cut downstream instability ~40%.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show and engagement scoring; improved recall **15-20%** on high-risk categories at precision >90%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | Hyderabad, India

Database & DevOps Performance Engineer (Intern) - Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** for the Synthesis Order-to-Cash platform; reduced query execution time ~20% and retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for version-controlled SQL deployments; reduced deployment errors >30% with structured release validation and rollback checkpoints.
- Designed performance dashboards (SQL Server DMVs, Grafana) tracking long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022 | Hyderabad, India

Machine Learning Engineer - Applied NLP, Video Analytics

- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised answer accuracy >85%.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Manga Lens - Real-Time AI Translation Chrome Extension (shipped, Chrome Web Store)

Full-stack MV3 extension integrating four AI vision providers (Claude, GPT-4o mini, GPT-4.1 Nano, Ollama). Multi-section capture pipeline, seven-day cache, per-domain configs for 29 sites, narrowed host permissions.

Dream Decoder - Multimodal AI Platform (FastAPI + React/TypeScript)

FastAPI + React/TS/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar, Replicate). Intermediate structured prompt transformation layers improved alignment ~30% and first-pass image success ~25-30%.

Agentic Healthcare Claims Processing & Fraud Risk System

Multi-agent pipeline with schema-validated JSON contracts between agents to prevent cascading hallucinations; RAG-grounded CPT/ICD validation; ANN similarity search for duplicate detection; audit-ready reasoning traces.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Backend: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, PHP

Cloud & Infra: Docker, Jenkins, CI/CD, RESTful APIs, cloud-ready deployment, Grafana, observability/logging, MLOps/LLMOps

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, embeddings, vector search, evaluation pipelines, guardrails

ML & Data: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, Pandas, NumPy, feature engineering

Work Authorization: F-1 OPT - authorized to work in the US now; future sponsorship required